

EXP NO:7B	DATA VISUALIZATION USING TABLEAU
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AIM:

To import, transform, analyze and visualize the Baseball Databank using Tableau Desktop and create interactive worksheets and dashboards for team, season, home-run and summary analysis.

ALGORITHM:

1. Connect the Baseball batting dataset to Tableau.
2. Check duplicates and missing values.
3. Create summary metrics for players, seasons, player-seasons, total runs and boundary/home-run-related measures used in the notebook.
4. Create a bar chart for the top teams by total runs.
5. Create a line chart for total runs by year.
6. Create a pie chart for home runs by team.
7. Create a team summary table.
8. Add suitable filters and calculated fields and analyze the final dashboard.

OUTPUT:

```
In [1]: from pathlib import Path

def find_data(filename):
    candidates = [
        Path(filename),
        Path("data") / filename,
        Path("../data") / filename,
        Path("../..data") / filename,
        Path("/content") / filename,
    ]
    for p in candidates:
        if p.exists():
            return p
    raise FileNotFoundError(f"{filename} not found. Upload the Baseball Databank CSV or keep it in the repository data folder.")

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv(find_data("Batting.csv"))
df["total_runs"] = df["R"]
df["boundary_runs"] = df["HR"] * 4

print("Dataset loaded successfully")
print("Shape:", df.shape)
display(df.head())
```

Dataset loaded successfully

Shape: (101332, 24)

	playerID	yearID	stint	teamID	lgID	G	AB	R	H	2B	3B	HR	RBI	SB	CS	BB	SO	IBB	HBP	SH	SF	GIDP	total_runs
0	abercda01	1871	1	TRO	NaN	1	4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	NaN	NaN	NaN	NaN	NaN	0.0
1	addybo01	1871	1	RC1	NaN	25	118.0	30.0	32.0	6.0	0.0	0.0	13.0	8.0	1.0	4.0	0.0	NaN	NaN	NaN	NaN	NaN	30.0
2	allisar01	1871	1	CL1	NaN	29	137.0	28.0	40.0	4.0	5.0	0.0	19.0	3.0	1.0	2.0	5.0	NaN	NaN	NaN	NaN	NaN	28.0
3	alliso01	1871	1	WS3	NaN	27	133.0	28.0	44.0	10.0	2.0	2.0	27.0	1.0	1.0	0.0	2.0	NaN	NaN	NaN	NaN	NaN	28.0

```
In [2]: print("Duplicate rows:", df.duplicated().sum())

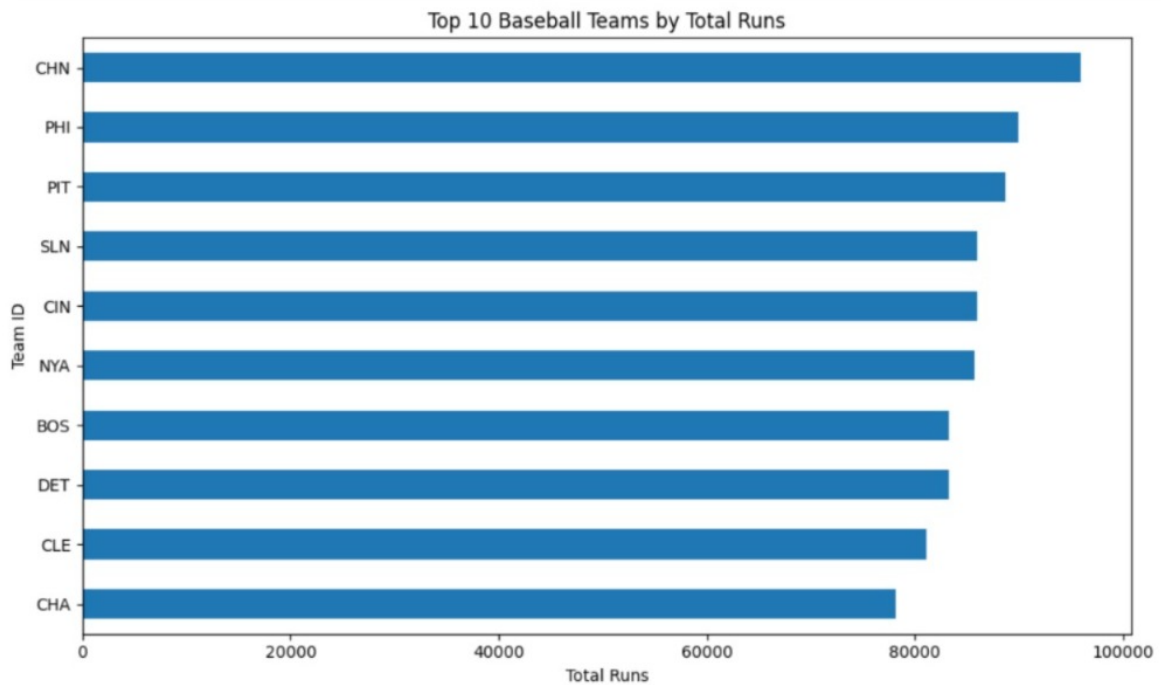
summary = pd.DataFrame({
    "Metric": ["Players", "Seasons", "Player-Seasons", "Total Runs", "Boundary Runs"],
    "Value": [df.playerID.nunique(), df.yearID.nunique(), len(df), int(df.total_runs.sum()), int(df.boundary_runs.sum())]
})
display(summary)
```

Duplicate rows: 0

	Metric	Value
0	Players	18659
1	Seasons	145
2	Player-Seasons	101332
3	Total Runs	1912795
4	Boundary Runs	1134692

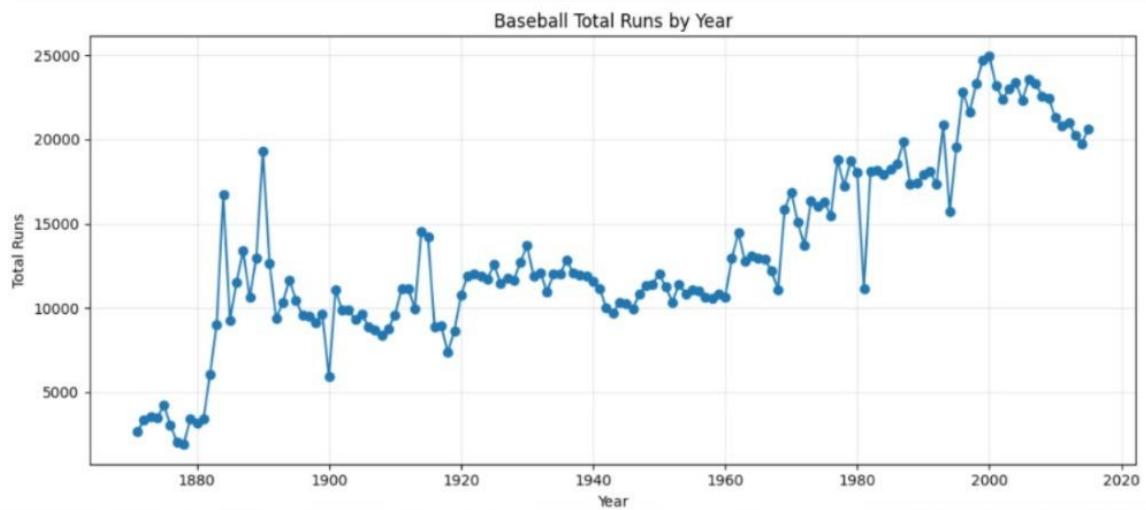
```
In [3]: team_runs = df.groupby("teamID")["total_runs"].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))
team_runs.sort_values().plot(kind="barh")
plt.title("Top 10 Baseball Teams by Total Runs")
plt.xlabel("Total Runs")
plt.ylabel("Team ID")
plt.tight_layout()
plt.savefig("Experiment_7B_Baseball_Team_Bar.png", dpi=150, bbox_inches="tight")
plt.show()
```



```
In [4]: season_runs = df.groupby("yearID")["total_runs"].sum()

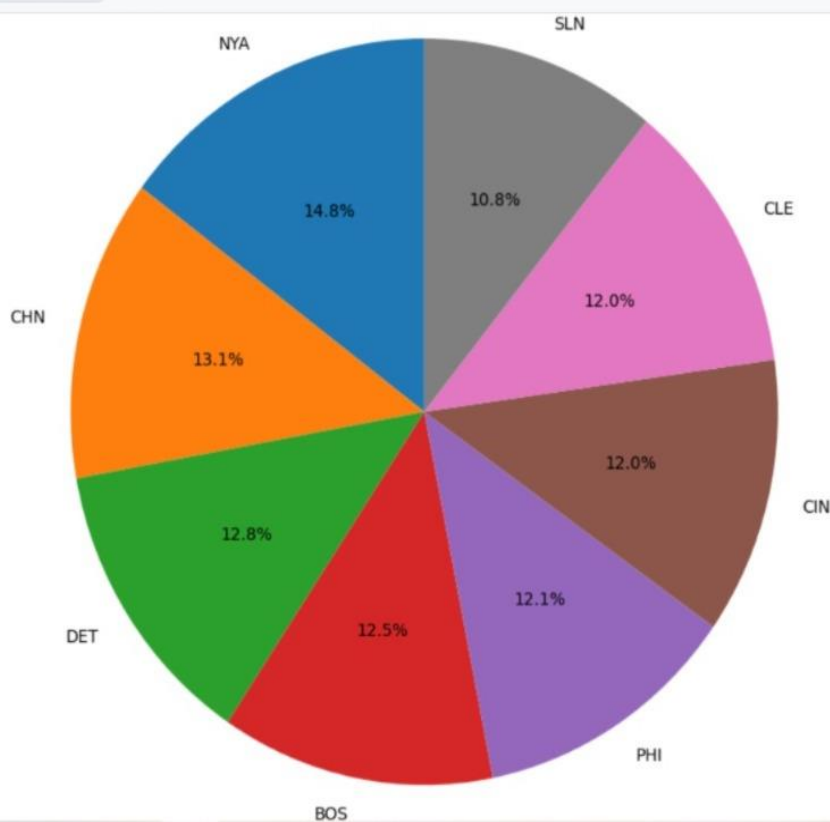
plt.figure(figsize=(11,5))
season_runs.plot(kind="line", marker="o")
plt.title("Baseball Total Runs by Year")
plt.xlabel("Year")
plt.ylabel("Total Runs")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("Experiment_7B_Baseball_Year_Line.png", dpi=150, bbox_inches="tight")
plt.show()
```



```
In [5]: top_hr = df.groupby("teamID")["HR"].sum().nlargest(8)

plt.figure(figsize=(8,8))
plt.pie(top_hr.values, labels=top_hr.index, autopct="%1.1f%%", startangle=90)
plt.title("Home Run Distribution Across Top Teams")
plt.tight_layout()
plt.savefig("Experiment_7B_Baseball_HomeRun_Pie.png", dpi=150, bbox_inches="tight")
plt.show()
```

Home Run Distribution Across Top Teams



```
In [6]: team_summary = df.groupby("teamID").agg(
    Total_Runs=("total_runs", "sum"),
    Boundary_Runs=("boundary_runs", "sum"),
    Player_Seasons=("playerID", "size")
).sort_values("Total_Runs", ascending=False).head(10)

display(team_summary)
```

teamID	Total_Runs	Boundary_Runs	Player_Seasons
CHN	96003.0	54120.0	4818
PHI	89959.0	50012.0	4721
PIT	88658.0	43512.0	4667
SLN	86003.0	44628.0	4627
CIN	85970.0	49532.0	4488
NYA	85715.0	60872.0	4220
BOS	83309.0	51532.0	4278
DET	83222.0	52640.0	4172
CLE	81116.0	49332.0	4451
CHA	78221.0	43168.0	4229

```
In [7]: print("Total Runs = SUM([total_runs])")
        print("Boundary Runs = SUM([boundary_runs])")
        print("Players = COUNTD([playerID])")
        print("Home Runs = SUM([HR])")
```

```
Total Runs = SUM([total_runs])
Boundary Runs = SUM([boundary_runs])
Players = COUNTD([playerID])
Home Runs = SUM([HR])
```

Result:

Thus, the Baseball Databank was successfully transformed, analyzed and visualized in Tableau using team, season, home-run and summary-table visualizations.